Fast forwarding conservative interacting transport by rate saturation

Proof candidate draft | 12 October 2026

Internally derived mathematical proof candidate. No novelty or priority claim, implementation, benchmark, external certification, or formal verification is asserted.

1 Actual computational task

Input: an explicitly supplied undirected graph on n vertices, m edges with nonnegative binary-rational exchange rates w_e, an initial permutation of n distinguishable labels (or an initial multicolor occupancy configuration), a nonnegative binary-rational horizon T, and accuracy 0<epsilon<=1. At rate w_{uv}, the labels at u and v exchange. This is the continuous-time interchange process. Coloring some labels occupied and others vacant gives symmetric simple exclusion, preserving both hard-core exclusion and particle conservation. Arbitrary static spatial disorder in edge rates is allowed, including exponentially separated scales.

Output: one full endpoint configuration whose law is within total-variation epsilon of the exact law at time T. The proposed algorithm is polynomial in n,m, the bit lengths of T and the rates, and 1/epsilon. It does not enumerate n! permutations or binomial(n,k) configurations and does not infer the full distribution from one-particle marginals.

This is an endpoint sampler, not a relative estimate of an exponentially rare event, a microscopic trajectory, a current-count sampler, a nonequilibrium interacting-potential model, or a reaction-network simulator. The physical assumption is unbiased, configuration-independent pair exchange with exclusion; rates are supplied physical inputs, not inferred for free. The input/output size is polynomial in the explicitly supplied graph, not in a succinctly encoded astronomically large lattice.

2 Claim uniformly cap rates above a polynomial threshold

Rescale time so the target horizon is 1, with rates r_e=T w_e. Let r_e^K=min(r_e,K). Write mu and mu_K for the endpoint laws from the same deterministic permutation under original and capped rates. Put c=n^2/2. For K>=2c,

```
||mu-mu_K||_TV <= (m c^2/K) [6+2 log(K/c)].                    (1)
```

The bound is uniform in every original rate, including rates far above K, and in the starting permutation. By projection it also holds for every multicolor/exclusion endpoint law. Choosing K=O((m n^4/epsilon) log(2mn/epsilon)) suffices; a deliberately loose explicit choice K=(12 m n^3/epsilon)^2 makes the cap contribution at most epsilon/4 for n>=2,m>=1. Indeed log(K/c)+3<=3 sqrt(K/c), so (1) is at most 6m c^(3/2)/sqrt(K)<=3m n^3/sqrt(K). T=0 and edgeless graphs are trivial.

Unlike clean-scale averaging, the argument does not require a gap between adjacent rate scales, and does not recursively enumerate or simulate equilibration regions.

3 The local discrepancy operation

Consider two interchange processes whose initial configurations differ only by exchanging the labels at the endpoints x,y of one edge e of current rate w>0.

Couple common edge clocks identically except when an edge joins the two discrepancy positions. At such an edge, use independent rate-w clocks for the two processes. A single jump then coalesces the configurations; afterwards use common clocks everywhere. Before coalescence, only two labels disagree. Their two positions evolve as independent continuous-time random walks on the weighted physical graph, killed when they meet. Every other label remains matched.

This coupling is an instance of the established interchange coupling described by Aldous and Fill, Chapter 14.5; the rate-weighted version follows from the same generator construction:
https://www.stat.berkeley.edu/~aldous/RWG/Book_Ralph/Ch14.S5.html

Let tau be the meeting time. The ordered pair chain before meeting has at most n(n-1) transient states and symmetric transition rates. At its initial state z=(x,y), there is direct killing at rate 2w, because either walker may cross e onto the other.

Let H be the positive killed generator and G=H^{-1} its Green matrix, restricted to the connected communicating pair domain. Equivalently G(z,u) is expected occupation time at u before killing when started at z. Symmetry gives G(z,u)=G(u,z). As a function of u, G(u,z) is harmonic off z and has zero absorbing boundary, so the maximum principle gives

```
0 <= G(z,u) <= G(z,z).
```

The direct conductance 2w from z to the absorbing boundary gives G(z,z)<=1/(2w). This follows either from effective resistance or by minimizing the Dirichlet energy with value 1 at z: the direct killing edge alone contributes 2w. Therefore

```
E_z tau = sum_u G(z,u) <= n^2/(2w) = c/w.
```

Consequently, the coupling inequality and Markov's inequality imply the edge-start smoothing bound

```
sup_sigma ||P_t(sigma,.)-P_t(sigma tau_e,.)||_TV
    <= a_w(t):=min(1,c/(wt)).                              (2)
```

There is no dependence on any other edge rate or on the number n! of configurations.

4 Group inversion supplies the missing second smoothing factor

Represent a configuration sigma as a map from positions to labels. The physical update is right multiplication sigma -> sigma tau_e. Let T_e f(sigma)=f(sigma tau_e), A_e=I-T_e, and

```
L=sum_e r_e A_e,    P_t=exp(-tL).
```

The transition kernel is K_t(sigma,eta)=p_t(sigma^{-1}eta), where p_t is the distribution from the identity. Reversibility with respect to uniform measure implies

```
p_t(g)=p_t(g^{-1}).
```

For any starting permutation sigma, the row gradient has entries

```
[delta_sigma P_t A_e](eta)
  =p_t(sigma^{-1}eta)-p_t(sigma^{-1}eta tau_e).
```

Changing variables g=sigma^{-1}eta and then g -> g^{-1}, its l1 norm equals

```
sum_g |p_t(g)-p_t(tau_e g)|
  =2 ||P_t(id,.)-P_t(tau_e,.)||_TV
  <=2 a_w(t).                                          (3)
```

Meanwhile (2), by duality, gives

```
||A_e P_t f||_infinity <=2 a_w(t) ||f||_infinity.          (4)
```

The distinction matters: for a general Markov chain an endpoint gradient is not the same as an initial-state gradient. The group convolution and inversion identity are the operation that makes both sides available here.

5 Two sided Duhamel sensitivity is integrable at infinite rate

Vary only edge e's rate w, fixing all other rates. Finite-state differentiation gives

```
dP_1/dw = -integral_0^1 P_s A_e P_{1-s} ds.
```

Because tau_e is an involution, A_e^2=2 A_e. Insert A_e=A_e^2/2. For ||f||_infinity<=1, (3)-(4) imply

```
|delta_sigma P_s A_e P_{1-s} f|
  <= (1/2)||delta_sigma P_s A_e||_1 ||A_e P_{1-s}f||_infinity
  <=2 a_w(s)a_w(1-s).
```

Total variation of a zero-mass signed measure is half its l1 norm, so

```
||d(delta_sigma P_1)/dw||_TV
  <= integral_0^1 a_w(s)a_w(1-s) ds.                    (5)
```

For q=c/w<=1/2, split the integral at q and 1-q:

```
integral_0^1 min(1,q/s)min(1,q/(1-s)) ds
  =-2q log(1-q)+2q^2 log((1-q)/q)
  <=q^2[4+2 log(1/q)].                                  (6)
```

A one-sided smoothing estimate would give only O(1/w), which cannot be integrated uniformly up to an arbitrarily huge rate. The two-sided factorization gives O(log w/w^2).

Integrating (5)-(6) from K to any rate W>=K gives

```
||Law_W(X_1)-Law_K(X_1)||_TV
  <= integral_K^infinity (c^2/w^2)[4+2log(w/c)] dw
  =(c^2/K)[6+2log(K/c)].                                (7)
```

The estimate is uniform in all other rates, so cap edges one by one and use the triangle inequality. This proves (1).

6 Constructive sampler and bit cost

Use K=(12 m n^3/epsilon)^2 and the following fully rational procedure:

1. Multiply input rates by T exactly and cap at K.
2. Let Lambda=sum_e r_e^K<=mK. If Lambda=0, return the initial state.
3. Choose the integer D=ceil(max(1,4 Lambda^2/epsilon)).
4. In each of D steps, independently do nothing with probability 1-Lambda/D, or select physical edge e with probability r_e^K/D and transpose its labels.
5. Return the final permutation or its occupancy/color projection.

D>=Lambda, so these are probabilities. Compare each step to the exact capped process over time 1/D. Its total clock count is Poisson(p), p=Lambda/D<=1. The total-variation distance between Poisson(p) and Bernoulli(p) is p(1-exp(-p))<=p^2. Conditional on a single event its edge is selected proportionally to r_e^K in both constructions. Coupling steps and ignoring multiple events gives endpoint error at most Dp^2=Lambda^2/D<=epsilon/4. Along with the cap error this leaves half the budget for finite random arithmetic.

All step probabilities are binary-rational quotients of polynomial-bit rationals; combine denominators, sample the corresponding integer categorical law by rejection from fair bits, and truncate each rejection routine with accumulated failure probability at most epsilon/2. A common denominator has at most the sum of the supplied denominator bit lengths. The number of steps is polynomial in n,m,1/epsilon and input bit lengths, independent of numerical T or numerical original rates. No real clock or special conditional sampler is assumed.

A more efficient implementation may use uniformization: draw N~Poisson(Lambda), then independently select each of N edges proportionally to r_e^K. Its expected swap count is Lambda<=mK. Truncating N at O(Lambda+log(1/epsilon)) and evaluating the Poisson masses to total error O(epsilon) gives a bounded polynomial-bit implementation. The deliberately conservative D-step scheme above suffices for the theorem and makes no practical runtime claim.

7 What this enables and what it does not

The output is a full interacting many-particle endpoint, including joint tracer positions and arbitrary occupancy correlations. With polynomially many independent samples, any bounded efficiently evaluable endpoint observable can be estimated to additive error. The gain is computational independence from the numerical observation time and from stiffness caused by enormous exchange-rate ratios. This differs from replacing transport by independent random walkers or computing only the mean density.

The mathematical model describes symmetric hard-core transport on a supplied network. It does not establish accurate material-specific exchange rates, permit biased/configuration-dependent exchange, or recover integrated currents and waiting-time history. Microscopic transition counts generally contain information deliberately discarded by capping. Parameter acquisition remains separate: slow rates relevant at time T can require absolute accuracy on the order of epsilon/T under elementary path coupling, while any rate already certified above K/T can be capped without learning its larger numerical value. Observing a very slow physical exchange may itself require a very long experiment. Rare-event relative error is also absent: total-variation epsilon permits losing events of probability much smaller than epsilon.

For observations at several prescribed times, a separate kernel approximation for each time increment, with errors summing to epsilon, gives joint snapshot sampling. That consequence does not reconstruct the intervening trajectory.

8 Primary comparator boundary

The basic discrepancy coupling is established, and uniformization is established. The present proof candidate combines the adjacent-edge meeting bound, convolution inversion, involution factorization, and two-sided Duhamel smoothing into a rate-saturation theorem. No claim of literature novelty is made.

- Aldous and Fill, Reversible Markov Chains and Random Walks on Graphs, Chapter 14.5, interchange coupling: https://www.stat.berkeley.edu/~aldous/RWG/Book_Ralph/Ch14.S5.html
- Van Dijk, van Brummelen, and Boucherie, Uniformization: Basics, extensions and applications, Performance Evaluation 118 (2018), 8-32: https://www.sciencedirect.com/science/article/pii/S0166531617301451
- Engblom, Strong convergence for split-step methods in stochastic jump kinetics, 2014 preprint: https://arxiv.org/abs/1412.6292 . This is a comparator for general stochastic splitting, not a source of the rate-capping guarantee.

The usual spectral-gap comparison for interchange is not needed in the argument above. Neither a gap-separated hierarchy nor a partition-function oracle is assumed.